

Alexander B. Chen

Shady Side Academy · Pittsburgh, PA · Class of 2027
27chena@shadysideacademy.org · Personal Website



Awards & Honors

Mathematics

United States of America Mathematics Olympiad (USAMO)[†] National · 2026

Scored a 5, top 50% of 250-300 participants was 6

American Invitational Mathematics Examination (AIME)* National · 2024/25/26

3-time qualifier; top score 12

American Mathematics Competition (AMC) 10/12 National · 2023/24/25

1st Place in Shady Side Academy for **AMC 12** in 2025-2026

1st Place in Shady Side Academy for **AMC 10** in 2023-2024 and 2024-25

	2025-26 (Grade 11)	2024-25 (Grade 10)	2023-24 (Grade 9)
AIME I*	12 (top 5% is 13)	8	5
AMC 10/12	117 (12A) 127.5 (12B, top 5%)	112.5 (10A, top 5%) 94.5 (12B)	115.5 (10A, top 5%) 111 (10B)

[†] First-ever **USAMO** qualifier from Shady Side Academy

* Highest-ever recorded score on the **AIME** from Shady Side Academy

MathLeague High School (Inter)National Championship

1st Place in **MathLeague High School National Championship**, Grade 11 National · 2026

2nd Place in **MathLeague High School International Championship**, Grade 10 International · 2025

MathLeague High School State Championship

1st Place in **MathLeague High School State Championship**, Grade 11 State · 2026

1st Place in **MathLeague High School State Championship**, Grade 10 State · 2025

United States of America Mathematics Talent Search (USAMTS)

Gold Prize (top 5%) National · 2026

The **USAMTS** is a free, national proof-based math competition (consisting of three rounds) open to all U.S. middle and high school students. It gives participants a month to write up detailed solutions to challenging problems and puzzles.

High School Mathematical Contest in Modeling (HiMCM)

Meritorious (M) prize (top 18%) International · 2025

Honorable Mention (HM) prize (top 45%) International · 2024

The **HiMCM**, run by COMAP, is an international team contest that trades numerical problem-solving for problem-designing. Teams of up to four are tasked with addressing a physical problem by designing a mathematical model and typing up a research-style paper.

- **2025 (Problem B).** A Pressure-State-Response model ranking Super Bowl host cities by environmental footprint was built that identified Tampa as the most sustainable host. I led the data analysis by using ARIMA for emissions forecasting, AHP-CRITIC for weighting, TOPSIS for city-ranking, and MICE and random-forest imputation for missing indicators, and we were able to generalize our results to the NBA All-Star Weekend.
- **2024 (Problem A).** A decision-tree expert system classifying sports into families and scoring each on popularity (Olympic YouTube view counts), accessibility, safety, and marketability was built that recommended which sports the International Olympic Committee should add or cut for the 2032 Brisbane Olympics. We used Monte Carlo simulation in Python to generate weighted inputs and logistic-growth curves to project the relevance of fast-emerging sports.

University of Pittsburgh High School Integration Bee (HSIB)

Champion, 1st Place Regional · 2026

2nd Place Regional · 2025

Business

Distributive Education Clubs of America (DECA)

DECA prepares high schoolers for careers in business, marketing, finance, and entrepreneurship, and runs a competition ladder from regionals up to the International Career Development Conference. My event was the Innovation Plan (EIP): a team starts from a blank page, invents a genuinely new product or service, works it into a full business plan, and defends it to a panel of judges.

International Career Development Conference (ICDC)

2-time qualifier International · 2024/25

State Career Development Conference (SCDC)

State Champion State · 2025
3rd Place State · 2024

- **2024 — CookGPT.** By curating a generative-AI cooking assistant, we attempted to tailor the experience to each user’s specific taste preferences. A single photograph of your fridge allows the app to identify the ingredients and the AI builds recipes from what’s on hand, tracks each meal’s nutrition to flag surpluses and shortfalls, and learns your tastes over time. The pitch was targeted to busy parents, seniors and the elderly, and people with dietary restrictions on a subscription model.
- **2025 — VITALIS.** A wearable health monitor was developed, our main target being aimed at strokes and heart attacks, where continuous biometric sensors feed a cloud database, machine-learning models flag danger patterns and raise preventive alerts, and an integrated microneedle system delivers emergency medication while contacting help. We also offer a plan to sell users’ data to hospitals and research institutions to both better understand patients’ medical conditions as well as making such data more widely accessible. The pitch was targeted to high-risk patients, clinicians, and research institutions on a device-plus-subscription model.

Music

Pittsburgh International Piano Competition (PIPC)

2nd Place. Invited to perform at the Winner’s Concert International · 2024

Steinway Society of Western Pennsylvania (SSWPA)

3-time winner. Invited to perform at Heinz Hall Regional · 2024/25/26

Westmoreland Symphony Orchestra (WSO): Young Artist Solo & Ensemble Festival

Gold Prize, scoring 98% Regional · 2026

Swimming

Eastern Zone Long Course Meters Age Group Championships (EZ LCM AGC)

Competed for the Allegheny Mountain Swimming (AMS) All-Star Swim Team with all states in the Eastern Zone. Zonal · 2024

Relays	13th Place in the 4×100m Freestyle Relay	13th Place in the 4×50m Freestyle Relay
	15th Place in the 4×100m Medley Relay	
Individual	100m Freestyle (58.06)	50m Freestyle (27.22)
	200m Individual Medley (IM) (2:28.35)	

Pennsylvania Interscholastic Athletic Association (PIAA) State Championships

2-time qualifier for the annual PIAA State Championship swim meet at Bucknell University. State · 2025/26
Placed 6th in the 4×100 Freestyle Relay (2025). Qualified for B-final in the 4×50 Freestyle Relay (2026). (short course yards, SCY)

Western Pennsylvania Interscholastic Athletic League (WPIAL)

3-time qualifier for annual WPIAL Championship swim meet at the University of Pittsburgh (short course yards, SCY).

	2025-26 (Grade 11)	2024-25 (Grade 10)	2023-24 (Grade 9)
4×50 Freestyle Relay	3rd	—	—
4×50 Medley Relay	4th	—	—
100 Backstroke	10th (57.80)	12th (58.55)	13th (59.73)
200 Freestyle	10th (1:51.07)	—	—
200 Individual Medley	—	15th (2:07.30)	—

Western Pennsylvania Interscholastic Athletic League (PIAA District VII) Section Champions

Shady Side Academy’s first-ever back-to-back undefeated seasons for Boys Varsity Swimming & Diving, earning back-to-back Section Titles/Section Champions. Regional · 2024-25/2025-26

Foreign Language

National Latin Exam (NLE)

Latin II, *Maxima Cum Laude*, Silver Medal; Latin III, *Cum Laude*, Honorable Mention

National · 2024/25

Pittsburgh Chinese School (PCS)

Outstanding Graduate; Chinese Cultural Scholarship

Regional · 2023-24

Graduated with honors after years of weekend coursework in Mandarin language and Chinese culture. Outstanding Graduate is awarded to two students per graduating class. Selected to give a speech at the graduation ceremony.

ACTIVITIES

Research & Projects

Program for Research in Mathematics (PReMa) — Texas A&M University

2025-present

A year of research on Anderson localization with Prof. Wencai Liu and Dr. Xueyin Wang, studying how disorder in a material confines quantum particles rather than letting them spread. Completed an intensive semester-long proof-based linear algebra course as preparation. Quantified the rate of exponential decay for the one-dimensional Anderson model with explicit error bounds, then used Monte Carlo simulation to estimate localization length across three types of disorder.

- **Publication.** *Finite-volume eigenfunction correlators for the one-dimensional Anderson model: a numerical study*, accepted by **Advances in Nonlinear Analysis** 2026
- Conferences and Talks
 - Accepted to present at **MAA MathFest** in Boston 2026
 - Talks presented at the **PReMa** research seminar and the **Ross Math Program** 2026

Projects & Writings

- **Roller-coaster dynamics: a day in Kennywood**, two data projects with Hunter Faderewski: we rode with accelerometers and extracted g-forces, airtime, and the shape of every drop. Slides about Kennywood's background: [slides](#). 2026
- A sharp piecewise generalization of **Nesbitt's inequality**. 2026
- **Biomimetic solar panel**, a Youth Design Challenge entry: fractal electrode layout generated from 725 leaf-cell epochs. 2025
- **The Political Faucet**, U.S. funds science like a faucet, opened in Cold War panic and shut in peacetime apathy. 2025
- **alecks67** (Substack), a blog for problems and half-finished ideas, including the **Cursed series** of deliberately perverse problems and constructions that look wrong and turn out to be right. 2025-present
- **An RC Car** that I 3D-printed along with Tony Chen through Fusion 360. A video demonstration [here](#). 2024

Expository Notes

Expository writing on topics I've found challenging and deeply interesting. All freely available.

- **Abstract Algebra & Number Theory**, a year-long course I designed and am teaching at Math Team meetings, with typeset lecture notes and problem sets. 2026-present
- **ARC** (AIME Roadmap & Curricula), a competition mathematics textbook series co-written with Ryan Gong. I wrote the **combinatorics volume**; number theory, geometry, and algebra volumes are in progress. 2025-present
- **Compiled lecture notes.** **Partial differential equations**, **set theory**, and **fractal geometry & chaos**. 2025-present
- **Multivariable Calculus**, a 360-page textbook written sophomore year, after my advisor asked why, if I disliked the assigned book, I didn't write my own. 2024-25

Summer Programs

Accepted to the Stanford University Mathematics Camp (**SUMaC online**), Program in Mathematics for Young Scientists (**PROMYS**), and the Ross Mathematics Program (**Ross**). I ended up participating in the Ross Math Program.

Ross Mathematics Program (Ross)

Summer 2026

Six weeks of intensive number theory in the **Ross/Ohio Camp** at the Otterbein University campus, rebuilding the subject from first principles: from ring and order axioms through unique factorization and quadratic reciprocity into group and field theory, and finally Galois theory. Completed 14 problem sets and took two advanced elective courses, in point-set topology and elliptic curves.

Accelerated High School Program (AHS) at the University of Pittsburgh

Enrolled at the University of Pittsburgh in the Kenneth P. Dietrich School of Arts and Sciences, taking university mathematics from multivariable calculus through partial differential equations alongside degree-seeking undergraduates, graded on the same basis.

- Introduction to Matrices & Linear Algebra (MA 0280) — A+ Summer 2024
- Analytic Geometry & Calculus 3 (MA 0240) — A+ Spring 2025
- Differential Equations (MA 0290) — A+ Spring 2026
- *Partial Differential Equations 1 (MA 1470) — in progress Fall 2026

* Upper-level coursework

Music

Carnegie Mellon Music Preparatory School (CMU Prep), Shady Side Academy String Orchestra

Piano since age 5; CMU Prep piano division since 2014; SSA String Orchestra pianist since 2023.

- **CMU Preparatory School.** Saturday music theory, music history and weekly lessons on the Carnegie Mellon campus 2014-present
under [Tatyana Gelman](#), performed in public recitals each semester (Winter, Spring, and occasionally Summer).
- **String Orchestra.** Sole pianist in the ensemble since Grade 9, which admits one student per year, under director 2023-present
Mr. Jeff Gross. Four concerts annually, monthly graded assignments, and honors-level credit each year past the first.

Swimming

Club Swimming: Jewish Community Center Sailfish (JCCS), Pittsburgh Elite Aquatics (PEAQ)

Joined the [JCCS](#) at age 7, and switched to [PEAQ](#) swim team at 15.

- **JCCS.** Achieved national AAA time, 2-time qualifier for the annual Eastern Zone LC Age Group Championships 2016-25
- **PEAQ.** Recently switched to the PEAQ team 2025-present

Varsity Swimming: Shady Side Academy (SSA)

3-time recipient of Varsity Letter (Freshman year, 2023-24, to Junior year, 2025-26)

2023-present

LEADERSHIP & VOLUNTEERING

Founder of Shady Side Academy Math Team & Math Club President

2023-present

Founded the school's first competition math program and team in 2025, securing \$6,000 in School Board funding in the 2025-2026 school year, recruiting and training students for the [AMC](#), [AIME](#), [MathLeague](#), and captaining the first-ever [ARML](#) team. Now in my third year as math club president, running twice-weekly meetings and weekly problem sessions, and olympiad preparation workshops, setting the curriculum, and teaching the club's first [abstract algebra](#) course. Membership grew from under 30 (5 active) to over 85, with 15-20 competing regularly, becoming one of the 3 largest clubs at the school.

Founder & Lead Organizer — Shady Side Academy Mathematics Olympiad (SSAMO)

2025-present

Created Shady Side's first student-led olympiad competition in 2025, drawing 20 registrations and 12 competing participants. Over eight months, I built and maintained the contest website, curated the problem set, and assembled the organizing team that handled logistics and 3D-printed the awards. I also designed a Problem of the Month page with full solutions and wrote a complete 25-problem mock AMC 12, and launched a Problem of the Month leaderboard in September 2026.

Pittsburgh Chinese School (PCS), 225 hours

Regional · 2024-present

2-time recipient of the Gold Service Award. Served as teaching assistant for both language and cultural math classes, 3 hours weekly across the school year, leading classroom discussions and readings, and grading assignments and quizzes. MCed the 2024 Spring Gala and graduation ceremony, was Youth Leader in the annual picnic, and organized canned-goods donations for a local food bank drive.

da Vinci Math Tutor & Teaching Assistant (TA), 20 hours

National · 2023-25

[da Vinci Math](#) is an online competition mathematics school serving students from all around the country. I led the weekly recitation and homework discussion, roughly an hour each week, being the TA for Introduction to Counting and Probability in 2023-24 and AP Calculus BC in 2024-25.

Greater Pittsburgh Community Food Bank, 9 hours

Regional · Summer 2025

Distributed food, fresh produce, and supplies to families facing food insecurity at local sites.